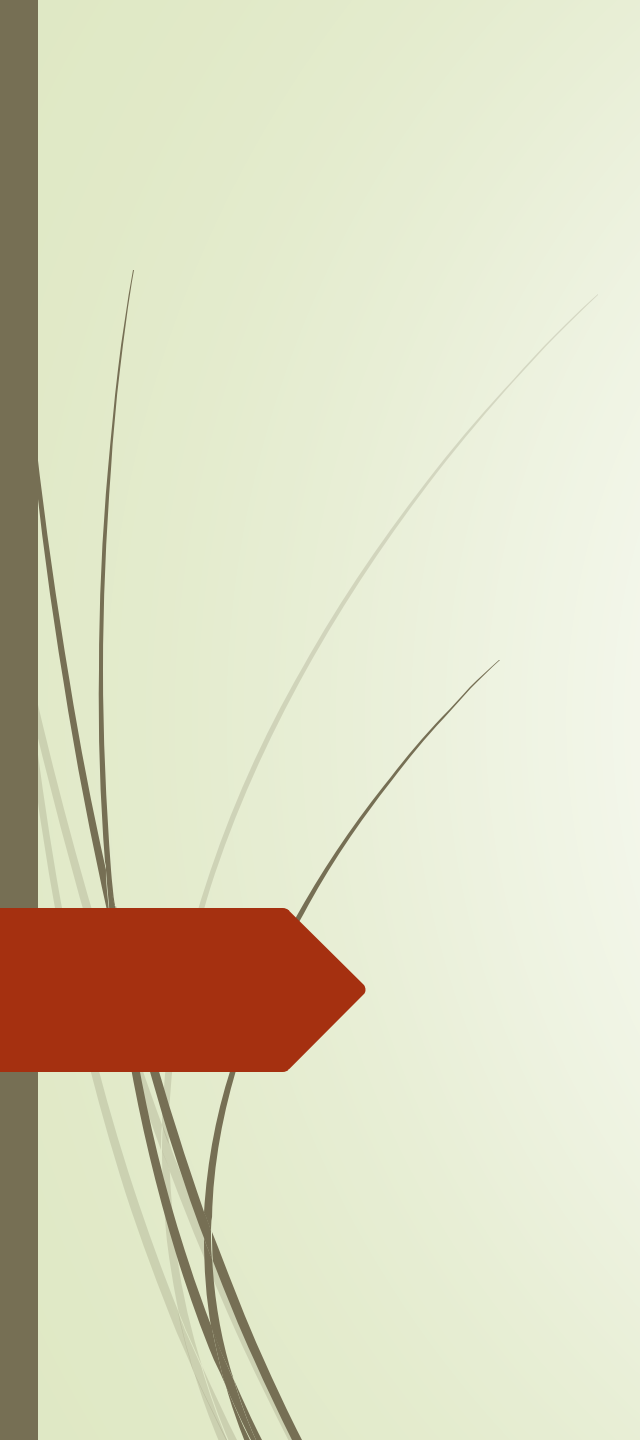




CT404-N OPERATING SYSTEM

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UNIT 1

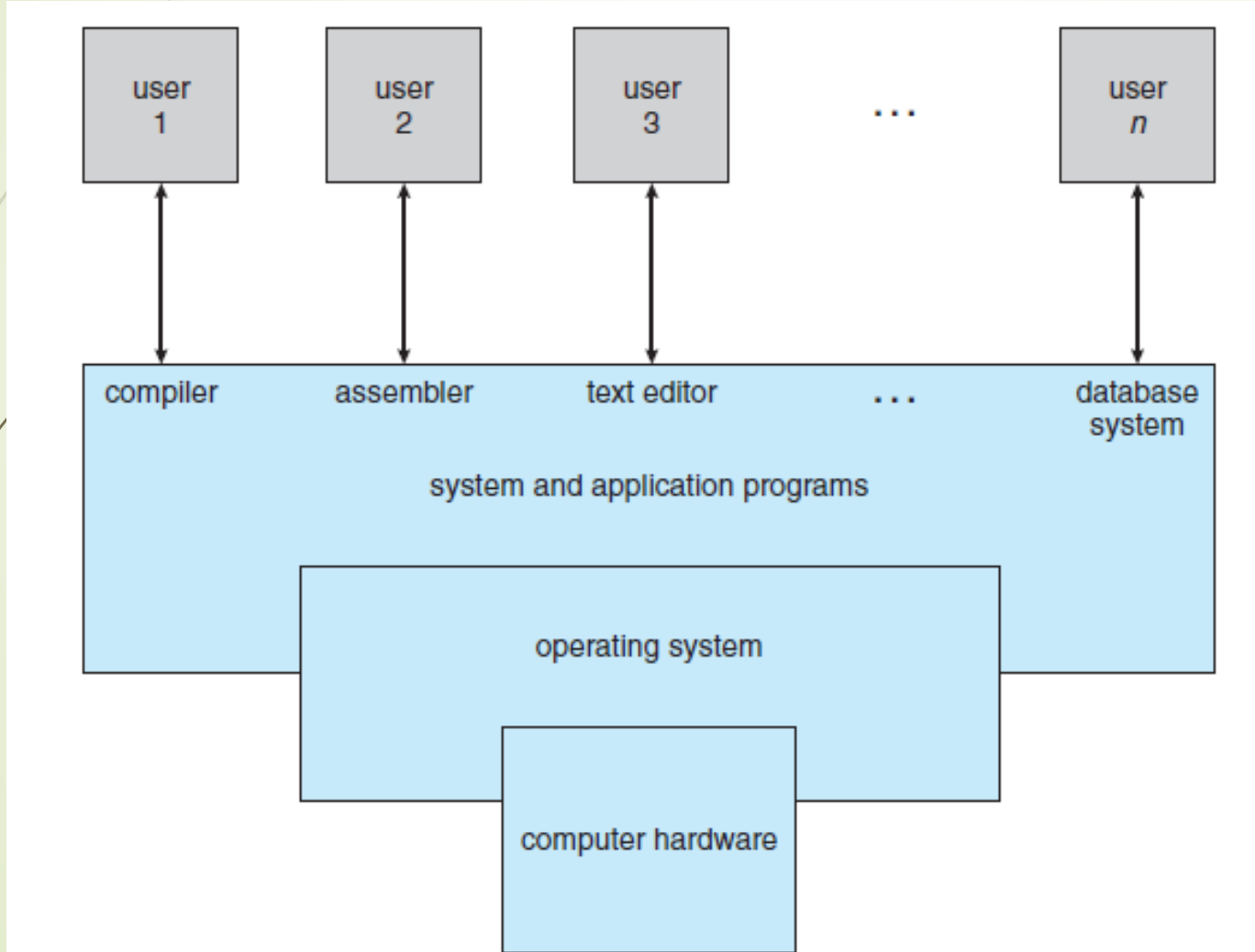
INTRODUCTION



OUTLINE

- ◄ Concept of OS
- ◄ Generations of OS
- ◄ Types of OS
- ◄ OS Services
- ◄ System Calls
- ◄ Structure of an OS
- ◄ Concept of Virtual Machine
- ◄ Case study on UNIX and WINDOWS OS

Components of a Computer System





←USER VIEW

- ← user's view of the computer varies according to the interface being used.
- ← users sit in front of a **PC**, a system is designed for one user to monopolize its resources.
- ← **ease of use**
- ← terminal connected to a **mainframe** or a **minicomputer**, such cases is designed to maximize resource utilization
- ← users sit at **workstations** connected to networks of other workstations and **servers**, designed to compromise between individual usability and resource utilization
- ← Smartphones, Most mobile computers are standalone units for individual users.



←SYSTEM VIEW

- ←an operating system as a **resource allocator**.
- ←A computer system has many resources that may be required to solve a problem: CPU time, memory space , file-storage space, I/O devices, and so on.
- ←the operating system must decide how to allocate them to specific programs and users so that it can operate the computer system efficiently
- ←An operating system is a control program. A **control program** manages the execution of user programs to prevent errors and improper use of the computer. It is especially concerned with the operation and control of I/O devices.



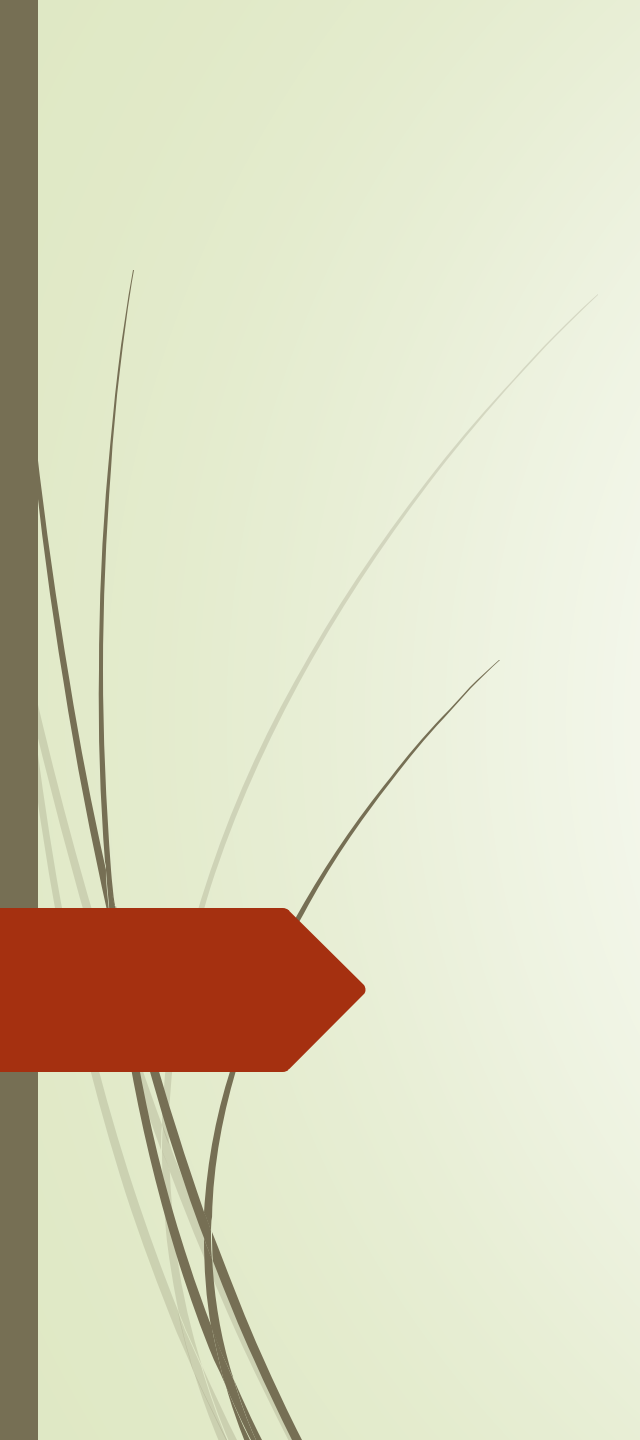
◄ Defining OS

- ◄ the one that we usually follow, is that the operating system is the one program running at all times on the computer—usually called the **kernel**.
- ◄ Along with the kernel, there are two other types of programs:
- ◄ **system programs**, which are associated with the operating system but are not necessarily part of the kernel, and
- ◄ application programs, which include all programs not associated with the operation of the system.

WHAT IS OS?

- An operating system is a program that manages the computer hardware.
- It also provides a basis for application programs and acts as an intermediary between the user and hardware.





GENERATIONS OF OS



First Generation (1940-1956) Vacuum Tubes

- used vacuum tubes for circuitry and [magnetic drums](#) for [memory](#)
- taking up entire rooms.
- very expensive to operate
- in addition to using a great deal of electricity, generated a lot of heat, which was often the cause of malfunctions.
- computers relied on [machine language](#)
- the lowest-level programming language understood by computers, to perform operations, and they could only solve one problem at a time
- Input was based on punched cards and paper tape, and output was displayed on printouts.
- The UNIVAC and [ENIAC](#) computers are examples .



Second Generation (1956-1963)

Transistors

- [Transistors](#) replaced vacuum tubes and ushered.
- The transistor was invented in 1947 but did not see widespread use in computers until the late 1950s.
- transistor was far superior to the vacuum tube, allowing computers to become smaller, faster, cheaper, more energy-efficient and more reliable than their first-generation predecessors.
- Though the transistor still generated a great deal of heat that subjected the computer to damage
- it was a vast improvement over the vacuum tube.
- second-generation computers still relied on punched cards for input and printouts for output.
- computers moved from cryptic [binary](#) machine language to symbolic, or [assembly](#), languages, which allowed programmers to specify instructions in words.
- [High-level programming languages](#) were also being developed at this time, such as early versions of [COBOL](#) and [FORTRAN](#).



Third Generation (1964-1971)

Integrated Circuits

- The development of the [integrated circuit](#) was the hallmark of the third generation of computers.
- Transistors were miniaturized and placed on [silicon chips](#), called [semiconductors](#).
- which drastically increased the speed and efficiency of computers.
- Instead of punched cards and printouts, users interacted with third generation computers through [keyboards](#) and [monitors](#)
- [interfaced](#) with an [operating system](#), which allowed the device to run many different [applications](#) at one time with a central program that monitored the memory.
- Computers for the first time became accessible to a mass audience because they were smaller and cheaper than their predecessors



Fourth Generation (1971-Present) Microprocessors

- The [microprocessor](#) brought the fourth generation of computers, as thousands of integrated circuits were built onto a single silicon chip.
- What in the first generation filled an entire room could now fit in the palm of the hand.
- The Intel 4004 chip, developed in 1971, located all the components of the computer—from the [central processing unit](#) and memory to input/output controls—on a single chip.
- As these small computers became more powerful, they could be linked together to form networks, which eventually led to the development of the Internet.
- Fourth generation computers also saw the development of [GUIs](#), the [mouse](#) and [handheld](#) devices



Fifth Generation (Present and Beyond) Artificial Intelligence

- Fifth generation computing devices, based on [artificial intelligence](#), are still in development.
- though there are some applications, such as [voice recognition](#), that are being used today.
- The use of [parallel processing](#) and superconductors is helping to make artificial intelligence a reality.
- [Quantum computation](#) and molecular and [nanotechnology](#) will radically change the face of computers in years to come.
- The goal of fifth-generation computing is to develop devices that respond to [natural language](#) input and are capable of learning and self-organization.



TYPES OF OS



Batch operating system

- The users of batch operating system do not interact with the computer directly.
- Each user prepares his job on an off-line device like punch cards and submits it to the computer operator.
- To speed up processing, jobs with similar needs are batched together and run as a group.
- Thus, the programmers left their programs with the operator.
- The operator then sorts programs into batches with similar requirements.



The problems with Batch Systems are following:

- Lack of interaction between the user and job.
- CPU is often idle, because the speeds of the mechanical I/O devices are slower than CPU.
- Difficult to provide the desired priority.



Multiprogramming operating system

- When two or more programs are residing in memory at the same time, then sharing the processor is referred to the multiprogramming.
 - Multiprogramming assumes a single shared processor.
 - Multiprogramming increases CPU utilization by organizing jobs so that the CPU always has one to execute.
- 



Multiprogramming operating system

- The operating system keeps several jobs in memory at a time.
- This set of jobs is a subset of the jobs kept in the job pool.
- The operating system picks and begins to execute one of the job in the memory.
- Multiprogramming operating system monitors the state of all active programs and system resources using memory management programs to ensures that the CPU is never idle unless there are no jobs



Multitasking operating system

- Multitasking refers to term where multiple jobs are executed by the CPU simultaneously by switching between them.
 - Switches occur so frequently that the users may interact with each program while it is running.
- 



Multitasking operating system

- The user gives instructions to the operating system or to a program directly, and receives an immediate response.
- Operating System handles multitasking in the way that it can handle multiple operations / executes multiple programs at a time.
- Multitasking Operating Systems are also known as Time-sharing systems.
- These Operating Systems were developed to provide interactive use of a computer system at a reasonable cost.
- A time-shared operating system uses concept of CPU scheduling and multiprogramming to provide each user with a small portion of a time-shared CPU.
- Each user has at least one separate program in memory.



...continued

- A program that is loaded into memory and is executing is commonly referred to as a process.
- When a process executes, it typically executes for only a very short time before it either finishes or needs to perform I/O.
- Since interactive I/O typically runs at people speeds, it may take a long time to complete. During this time a CPU can be utilized by another process.
- Operating system allows the users to share the computer simultaneously. Since each action or command in a time-shared system tends to be short, only a little CPU time is needed for each user.
- As the system switches CPU rapidly from one user/program to the next, each user is given the impression that he/she has his/her own CPU, whereas actually one CPU is being shared among many users.



Time-sharing operating systems

- Time sharing is a technique which enables many people, located at various terminals, to use a particular computer system at the same time.
- Time-sharing or multitasking is a logical extension of multiprogramming.
- Processor's time which is shared among multiple users simultaneously is termed as time-sharing.
- The main difference between Multiprogrammed Batch Systems and Time-Sharing Systems is that in case of multiprogrammed batch systems, objective is to maximize processor use, whereas in Time-Sharing Systems objective is to minimize response time.



Time-sharing operating systems

- Multiple jobs are executed by the CPU by switching between them, but the switches occur so frequently.
- Thus, the user can receive an immediate response. For example, in a transaction processing, processor execute each user program in a short burst or quantum of computation.
- That is if n users are present, each user can get time quantum. When the user submits the command, the response time is in few seconds at most.
- Operating system uses CPU scheduling and multiprogramming to provide each user with a small portion of a time.
- Computer systems that were designed primarily as batch systems have been modified to time-sharing systems.



Real Time operating System

- Real time system is defines as a data processing system in which the time interval required to process and respond to inputs is so small that it controls the environment.
 - Real time processing is always on line whereas on line system need not be real time.
 - The time taken by the system to respond to an input and display of required updated information is termed as response time.
 - So in this method response time is very less as compared to the online processing.
- 



Real Time operating System

- Real-time systems are used when there are rigid time requirements on the operation of a processor or the flow of data and real-time systems can be used as a control device in a dedicated application.
- Real-time operating system has well-defined, fixed time constraints otherwise system will fail.
- For example Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, and home-appliance controllers, Air traffic control system etc.



Types of Real Time operating System

- There are two types of real time os:

1. Hard real-time systems

Hard real-time systems guarantee that critical tasks complete on time.

In hard real-time systems secondary storage is limited or missing with data stored in ROM. In these systems virtual memory is almost never found.

2. Soft real-time systems

Soft real time systems are less restrictive. Critical real-time task gets priority over other tasks and retains the priority until it completes.

Soft real-time systems have limited utility than hard real-time systems. For example, Multimedia, virtual reality, Advanced Scientific Projects like undersea exploration and planetary rovers etc.



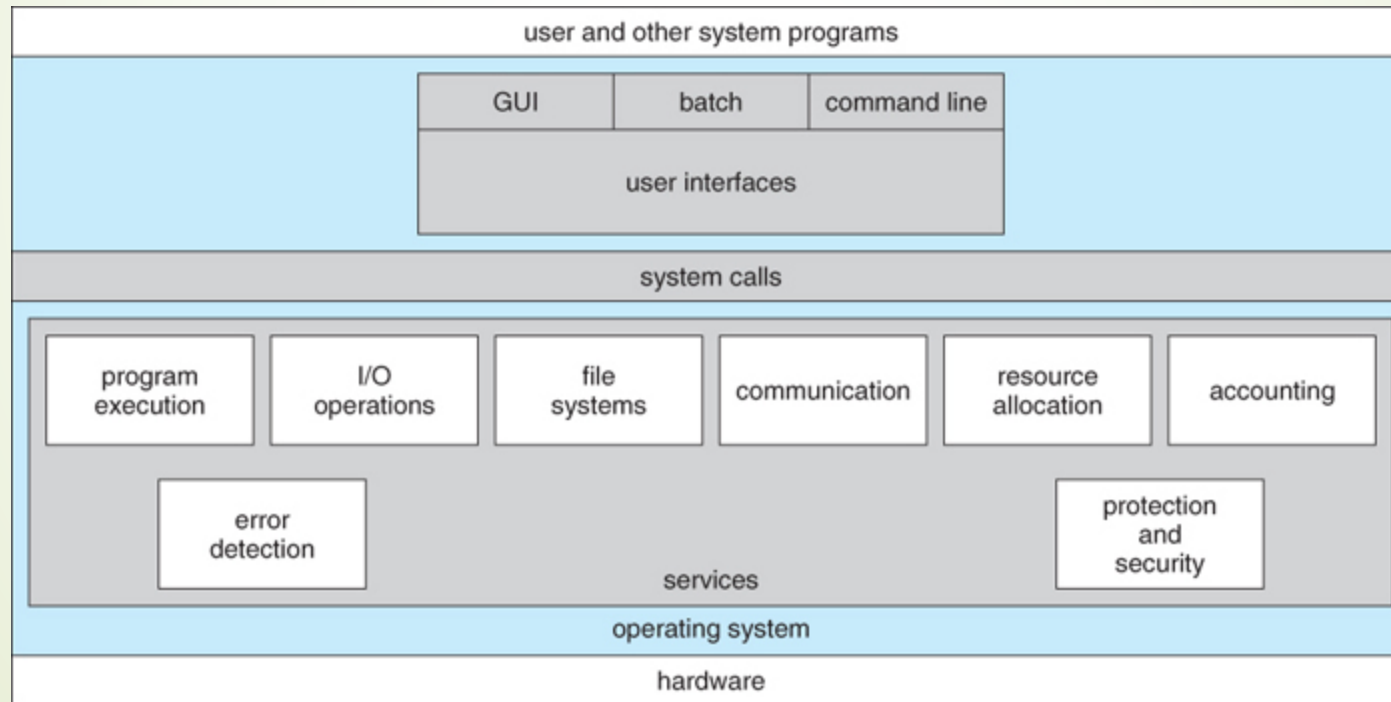
Distributed operating System

- Distributed systems use multiple central processors to serve multiple real time application and multiple users.
- Data processing jobs are distributed among the processors accordingly to which one can perform each job most efficiently.
- The processors communicate with one another through various communication lines (such as high-speed buses or telephone lines).
- These are referred as loosely coupled systems or distributed systems. Processors in a distributed system may vary in size and function.
- These processors are referred as sites, nodes, and computers and so on.



OS SERVICES

The operating system provides the programming environment in which a programmer works on a computer system. The user program requests various resources through the operating system. The operating system gives several services to utility programmers and users. Applications access these services through application programming interfaces or system calls. By invoking those interfaces, the application can request a service from the operating system, pass parameters, and acquire the operation outcomes.





Following are the services provided by an operating system -

- Program execution
- Control Input/output devices
- Program creation
- Error Detection and Response
- Accounting
- Security and Protection
- File Management
- Communication



Program execution

To execute a program, several tasks need to be performed. Both the instructions and data must be loaded into the main memory. In addition, input-output devices and files should be initialized, and other resources must be prepared. The Operating structures handle these kinds of tasks. The user now no longer should fear the reminiscence allocation or multitasking or anything.

Control Input/output devices

As there are numerous types of I/O devices within the computer system, and each I/O device calls for its own precise set of instructions for the operation. The Operating System hides that info with the aid of presenting a uniform interface. Thus, it is convenient for programmers to access such devices easily.

Program Creation

The Operating system offers the structures and tools, including editors and debuggers, to help the programmer create, modify, and debugging programs.



Error Detection and Response

An Error in a device may also cause malfunctioning of the entire device. These include hardware and software errors such as device failure, memory error, division by zero, attempts to access forbidden memory locations, etc. To avoid error, the operating system monitors the system for detecting errors and takes suitable action with at least impact on running applications.

While working with computers, errors may occur quite often. Errors may occur in the:

- **Input/ Output devices:** For example, connection failure in the network, lack of paper in the printer, etc.
- **User program:** For example: attempt to access illegal memory locations, divide by zero, use too much CPU time, etc.
- **Memory hardware:** For example, Memory error, the memory becomes full, etc.

To handle these errors and other types of possible errors, the operating system takes appropriate action and generates messages to ensure correct and consistent computing.

Accounting

An Operating device collects utilization records for numerous assets and tracks the overall performance parameters and responsive time to enhance overall performance. These personal records are beneficial for additional upgrades and tuning the device to enhance overall performance.

Security and Protection

Operating device affords safety to the statistics and packages of a person and protects any interference from unauthorized users. The safety feature counters threats, which are published via way of individuals out of doors the manage of the running device.



File management

Computers keep data and information on secondary storage devices like magnetic tape, magnetic disk, optical disk, etc. Each storage media has its capabilities like speed, capacity, data transfer rate, and data access methods.

For file management, the operating system must know the types of different files and the characteristics of different storage devices. It has to offer the protection and safety mechanism of documents additionally.

Communication

The operating system manages the exchange of data and programs among different computers connected over a network. This communication is accomplished using message passing and shared memory.



SYSTEM CALL



System Calls

System calls provide an interface to the services made available by an operating system. These calls are generally available as routines written in C and C++, although certain low-level tasks

What are System Calls

- System Calls provide the Interface between a process and the Operating System.
- These calls are generally available as Assembly language instruction.
- System Calls can also be made directly through HLL programs for certain systems.
- UNIX System calls can be invoked directly from a C or C++ program.



Scenario of System Calls

An Example of how system calls are used.

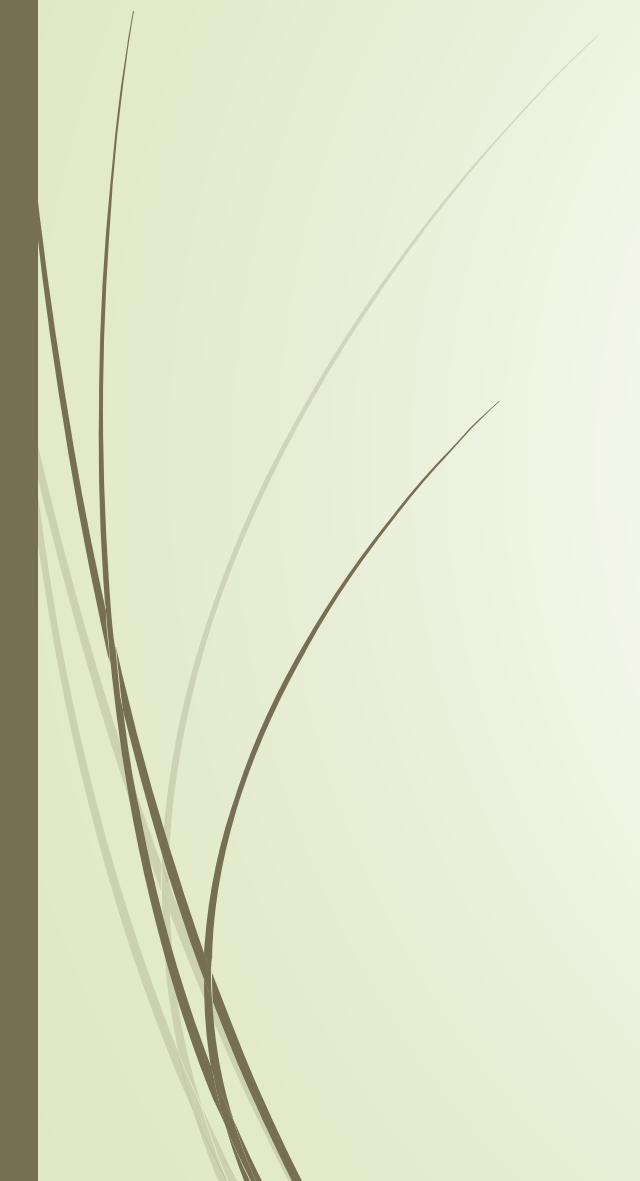
- Consider a simple program to copy data from one file to another file .
- The first input of the program will be the name of both files.
- These names can be specified depending on the design of Operation System
- Once the file names are obtained, the program opens an input file and create an output file
- Each of this operations require other System calls and may encounter possible error conditions.

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System Calls

- System calls provide the interface between a process and the OS:
 - ☞ Generally available as assembly-language instructions.
 - ☞ Languages defined to replace assembly language for systems programming allow system calls to be made directly (e.g., C, C++).
- A system call is used to complete a specific task:
 - ☞ Reading input file name
 - ☞ Opening a file
 - ☞ Error message if any
 - ☞ Abnormal termination
 - ☞ Deletion of a file
 - ☞ Prompting a message
 - ☞ Reading data for the file
 - ☞ Writing data to the file or console
 - ☞ Closing the file
 - ☞ Normal termination
- Three general methods are used to pass parameters between a running program and the operating system:
 - ☞ Pass parameters in *registers*.
 - ☞ Store the parameters in a table in memory, and the table address is passed as a parameter in a register (used in linux)
 - ☞ *Push* (store) the parameters onto the *stack* by the program, and *pop* off the stack by OS.



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graph TD; UP[User Programs] -- "System Calls" --> K[Kernel]; K --> H[Hardware];
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User Programs

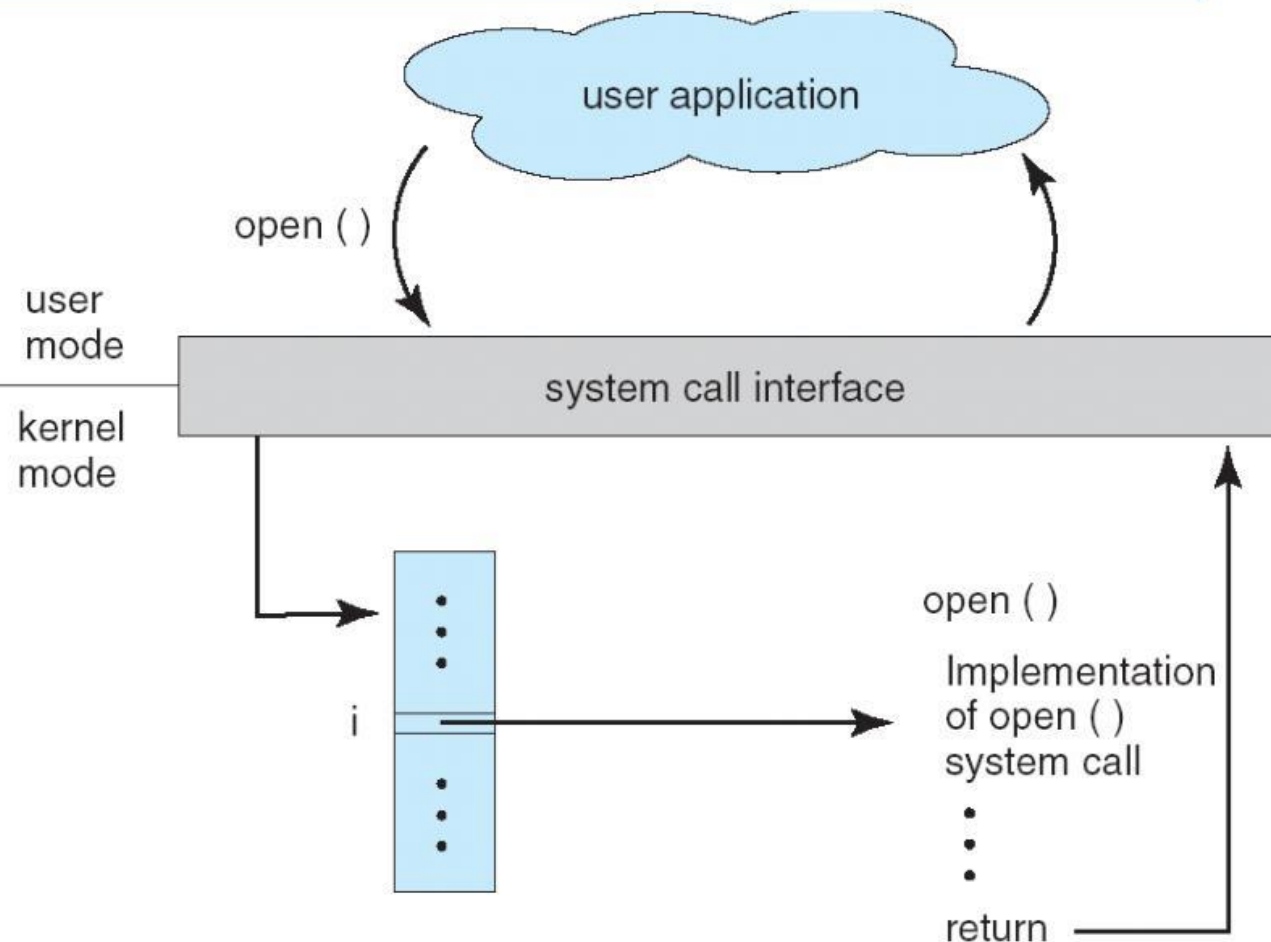
System Calls

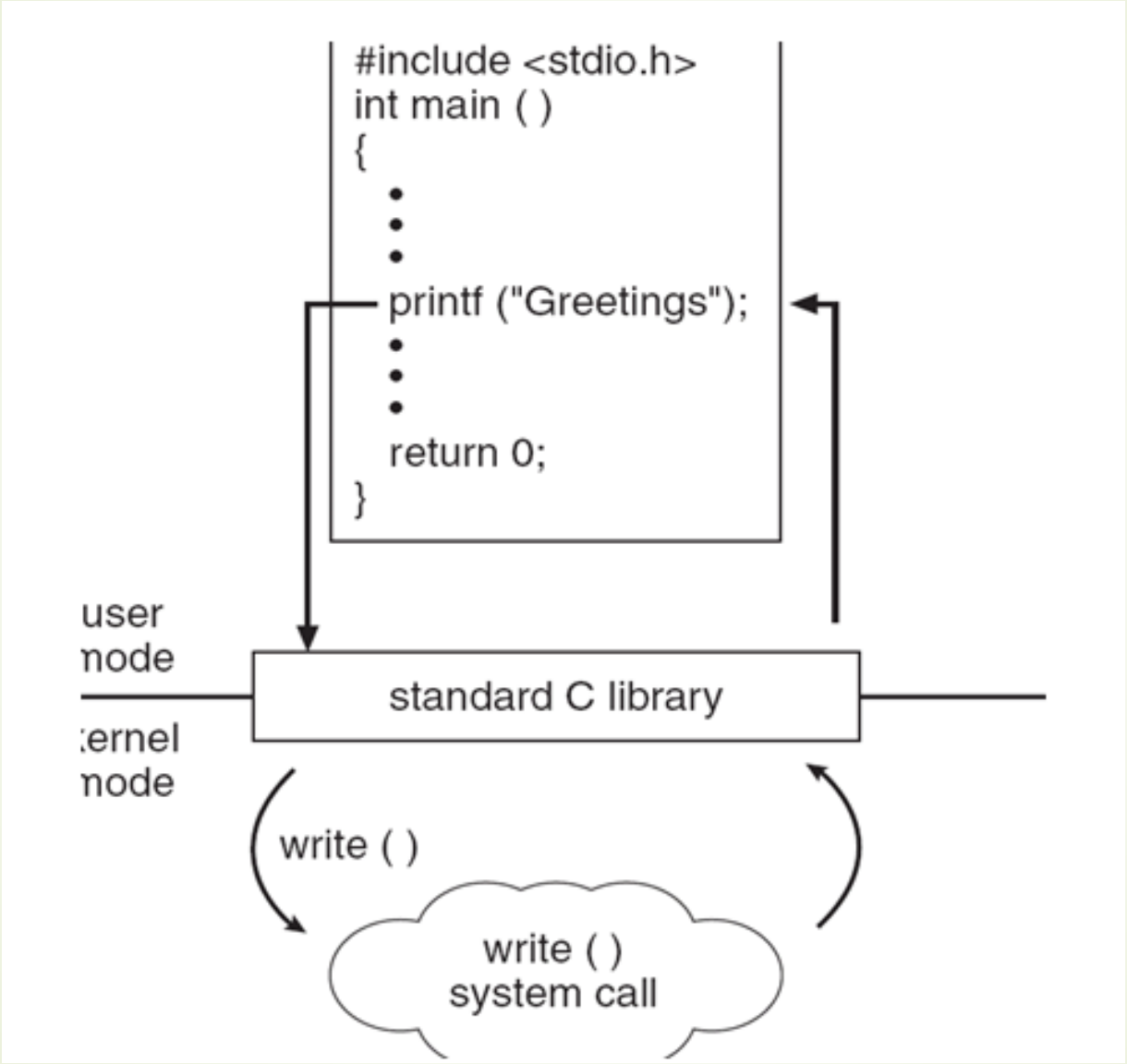
Kernel

Hardware

Difference
Between.com

API – System Call – OS Relationship





Types of System Calls

■ Process control

- ☞ End, Abort
- ☞ Load, Execute
- ☞ Create process, Terminate process
- ☞ Get process attributes, Set process attributes
- ☞ Wait for time
- ☞ Wait event, Signal event
- ☞ Allocate memory, Free memory

■ File management

- ☞ Create file, Delete file
- ☞ Open, Close
- ☞ Read, Write, Reposition
- ☞ Get file attributes, Set file attributes

■ Device management

- ☞ Request device, Release device
- ☞ Read, Write, Reposition
- ☞ Get device attributes, Set device attributes
- ☞ Logically attach devices, Logically detach devices



Types of System Calls

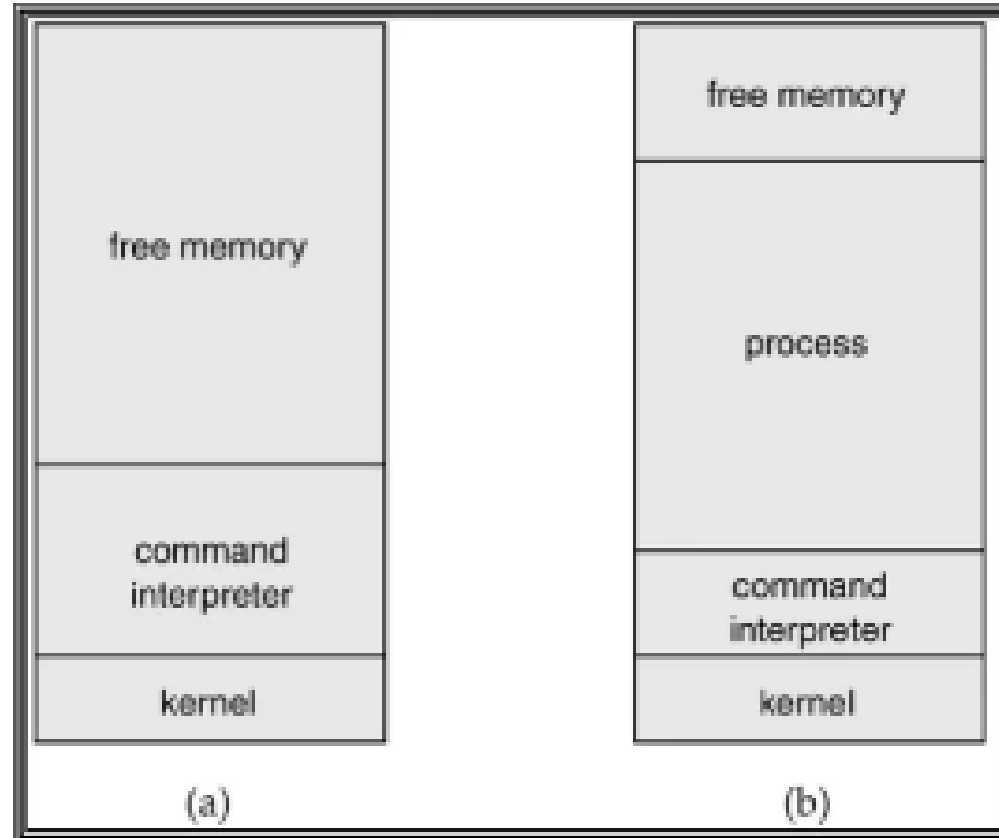
■ Information maintenance

- ☞ Get time or date, Set time or date
- ☞ Get system data, Set system data
- ☞ Get process, file, or device attributes, Set process, file, or device attributes

■ Communications

- ☞ Create communication connection, Delete communication connection
- ☞ Send messages, Receive messages
- ☞ Transfer status information
- ☞ Attach remote devices, Detach remote devices

MS-DOS Execution



At System Start-up

Running a Program



OS STRUCTURE

What is a kernel ?

- The kernel is a computer program at the core of a computer's operating system and has complete control over everything in the system. It manages the operations of the computer and the hardware.
- **There are five types of kernels :**
 - A micro kernel, which only contains basic functionality;
 - A monolithic kernel, which contains many device drivers.
 - Hybrid Kernel
 - Exo kernel
 - Nano kernel

- **Microkernel** :

kernel manages the operations of the computer, In microkernel the user services and kernel services are implemented in different address space. The user services are kept in user address space, and kernel services are kept under kernel address space.

- **Monolithic kernel** :

In Monolithic kernel, the entire operating system runs as a single program in kernel mode. The user services and kernel services are implemented in same address space.



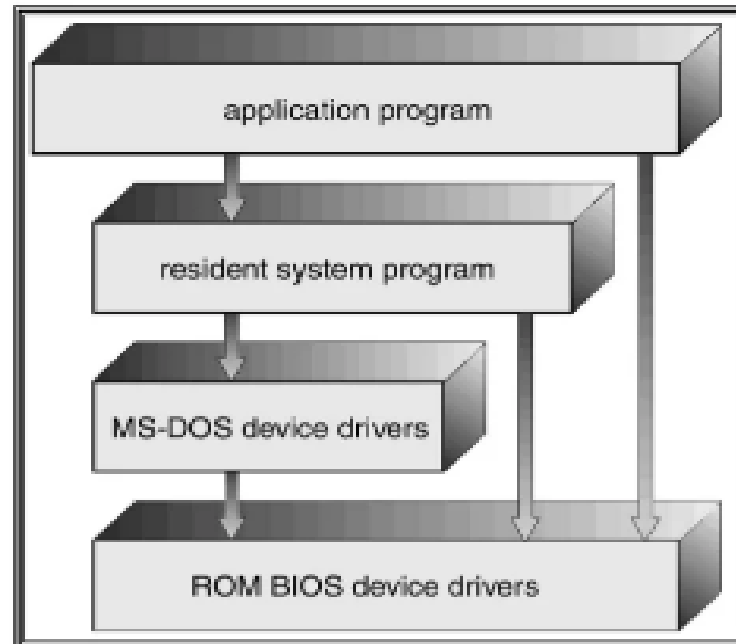
A common approach is to partition the task into small components, or modules, rather than have one **monolithic** system.

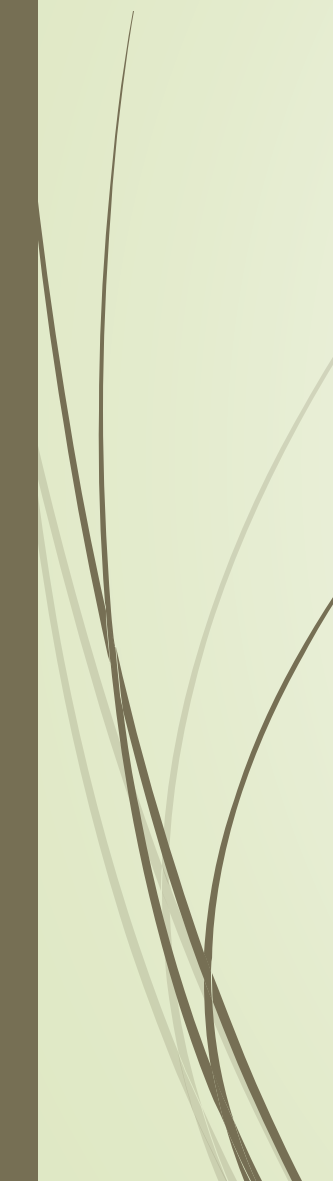
- Each of these modules should be a well-defined portion of the system, with carefully defined inputs, outputs, and functions.

Simple Structure

Simple System Structure (MS-DOS)

- MS-DOS was written to provide the most functionality in the least space.
- It was not divided into modules carefully.
- Although MS-DOS has some structure, its interfaces and levels of functionality are not well separated.



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- Many operating systems do not have well-defined structures.
 - Frequently, such systems started as small, simple, and limited systems and then grew beyond their original scope. MS-DOS is an example of such a system.
 - In MS-DOS, the interfaces and levels of functionality are not well separated.
 - For instance, application programs are able to access the basic I/O routines to write directly to the display and disk drives.
 - Such freedom leaves MS-DOS vulnerable to errant (or malicious) programs, causing entire system crashes when user programs fail.

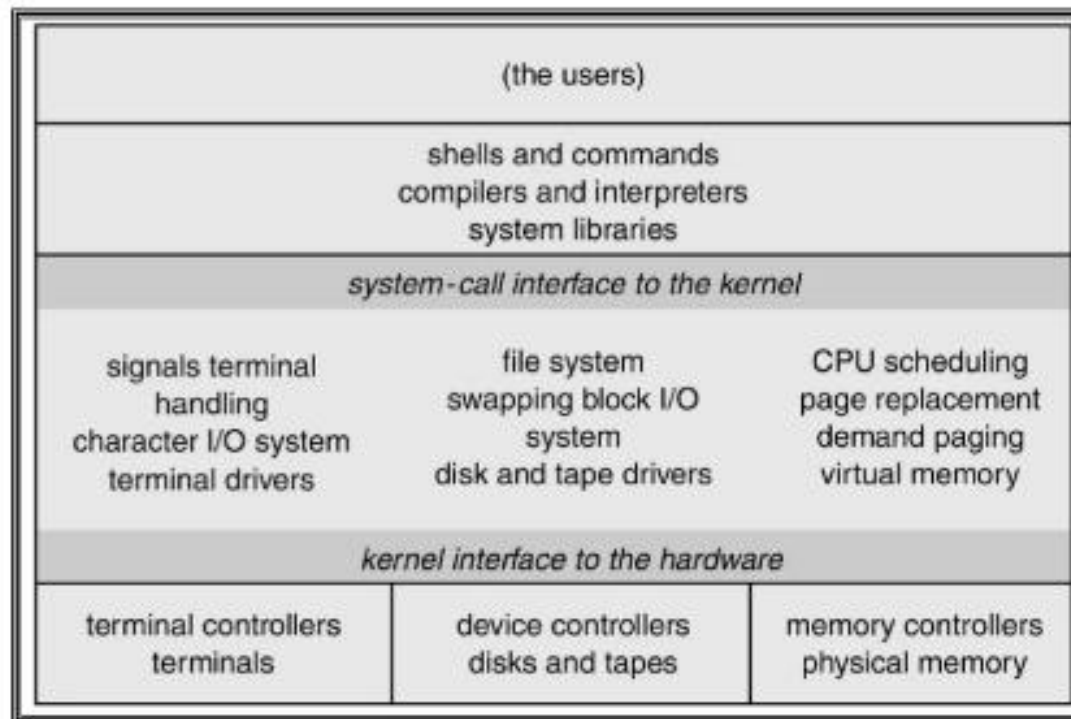


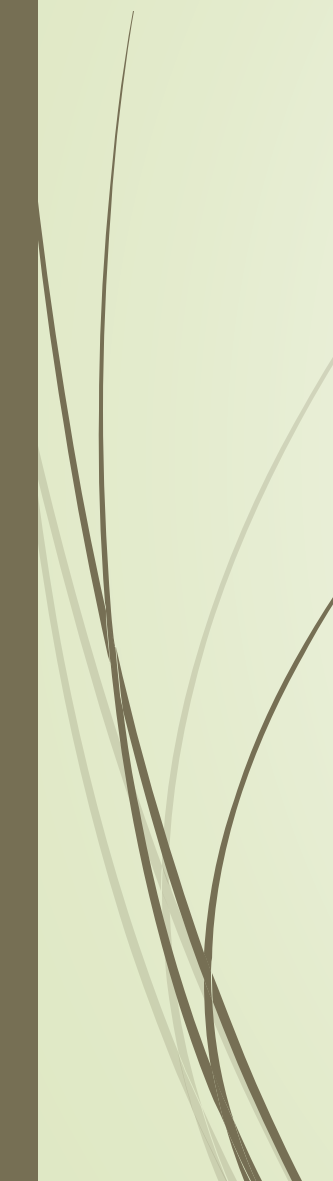
Simple System Structure (UNIX)

- UNIX – limited by hardware functionality, the original UNIX operating system had limited structuring. The UNIX OS consists of two separable parts:
 - ✦ **Systems programs**
 - ▢ Systems programs are provided to define the user interface.
 - ✦ **The kernel**
 - ▢ Consists of everything below the system-call interface and above the physical hardware.
 - ▢ Separated into a series of interfaces and device drivers.
 - ▢ Different components (file system, CPU scheduling, memory management and other OS functions) were layered in its design.
 - ✦ OS may provide smaller modules for much greater control of computer resources, incorporating changes in the separate modules easily to enhance functionality and hide information in different components.

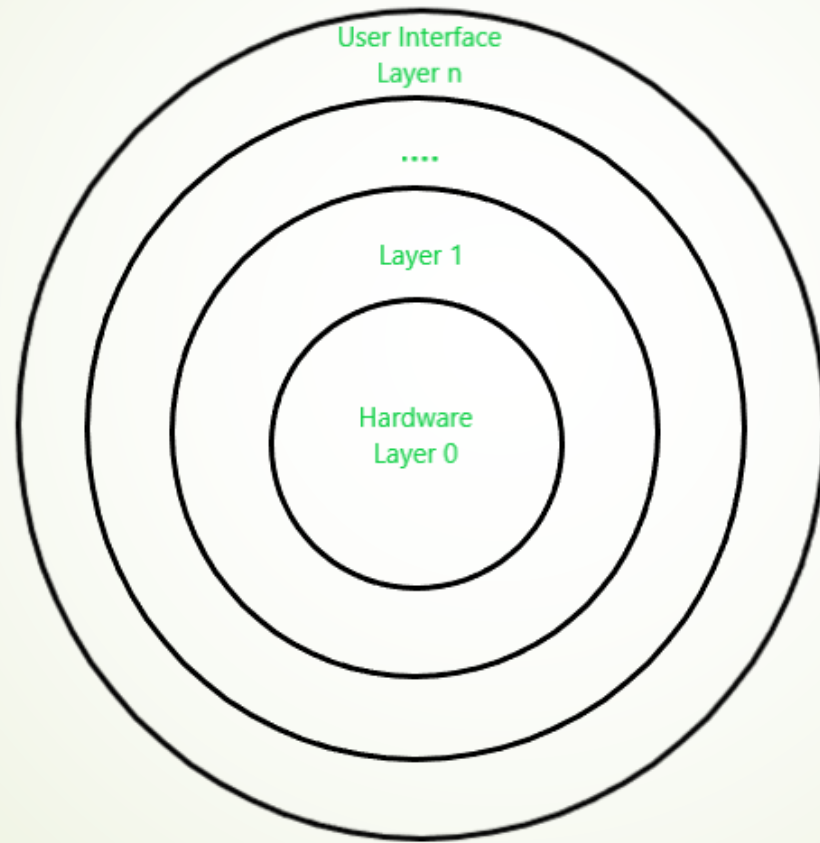
Monolithic Structure

UNIX System Structure



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- 
- Another example of limited structuring is the original UNIX operating system. Like MS-DOS, UNIX initially was limited by hardware functionality.
 - It consists of two separable parts: the kernel and the system programs.
 - Kernel is further separated into a series of interfaces and device drivers, which have been added and expanded over the years as UNIX has evolved.
 - We can view the traditional UNIX operating system as being layered to some extent
 - Everything below the system-call interface and above the physical hardware is the kernel.
 - The kernel provides the file system, CPU scheduling, memory management, and other operating-system functions through system calls.
 - Taken in sum, that is an enormous amount of functionality to be combined into one level.
 - This monolithic structure was difficult to implement and maintain.

Layered OS





Layered Approach

- The operating system is divided into a number of layers (levels), each built on top of lower layers. The bottom layer (layer 0) is the hardware; the highest (layer N) is the user interface.
- With modularity, layers are selected such that each uses functions (operations) and services of only lower-level layers.
- OS has a greater control over the computer resources and applications that make use of it.
- Provides more freedom to the implementers to make changes in the inner workings of the system.
- Provides information hiding by allowing the programmers to implement the low-level routines using top-down modular programming approach.
- Layered approach is used for modularization of the OS.
- Higher-level layers can invoke lower-level layers.
- Less number of layers provide more functionality at each level while reducing the problems.



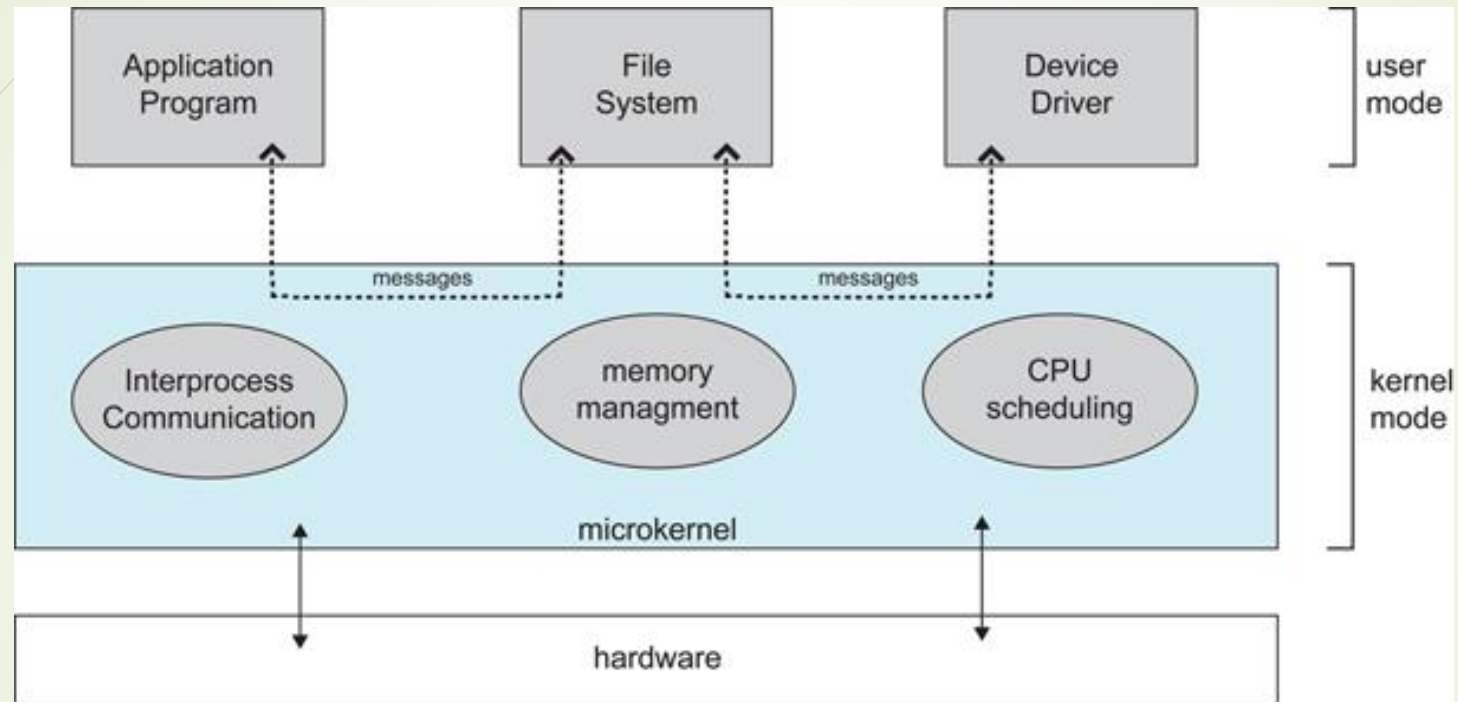
Layered Approach

- **Advantages:** modularization makes debugging and verification much easier; design and implementation is simplified; each layer hides certain data structures, operations and hardware details from higher-level layers.
- **Disadvantages:** level definition and implementation is very difficult; order of levels for the design of layered structure is problematical; interaction between layers and passing of parameters / data is very complex; less efficient as a system call takes longer time for execution than a non-layered system.
- **Examples:** Technische Hogeschool Eindhoven (THE) OS and Venus layer structure.



Microkernels

- UNIX expanded, the kernel became large and difficult to manage. In the mid-1980s, researchers at Carnegie Mellon University developed an operating system called **Mach** that modularized the kernel using the **microkernel** approach.
- This method structures the operating system by removing all nonessential components from the kernel and implementing them as system and user-level programs.
- The result is a smaller kernel.

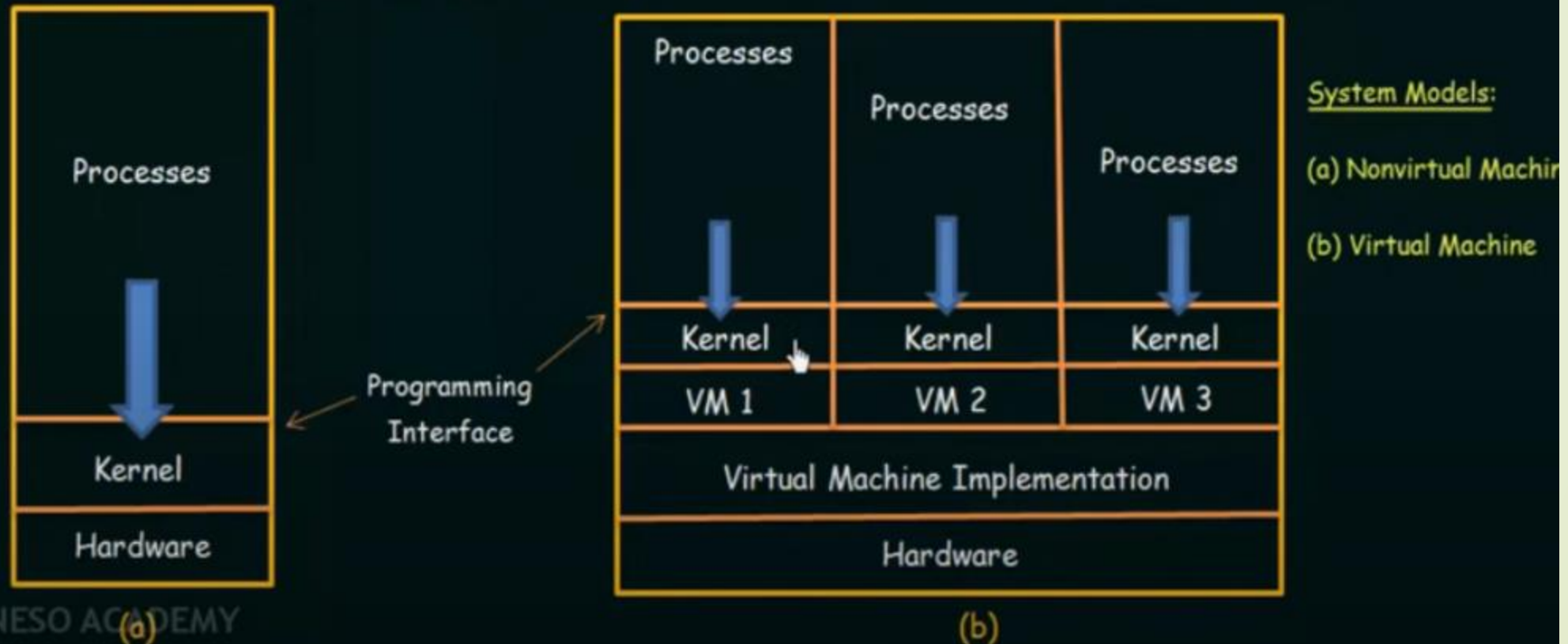


- 
- The main function of the microkernel is to provide communication between
 - the client program and the various services that are also running in user space.
 - Communication is provided through **message passing**
- 



Virtual Machines

The fundamental idea behind a virtual machine is to abstract the hardware of a single computer (the CPU, memory, disk drives, network interface cards, and so forth) into several different execution environments, thereby creating the illusion that each separate execution environment is running its own private computer.



IMPLEMENTATION

Virtual Machine Software -
Virtual Machine itself -

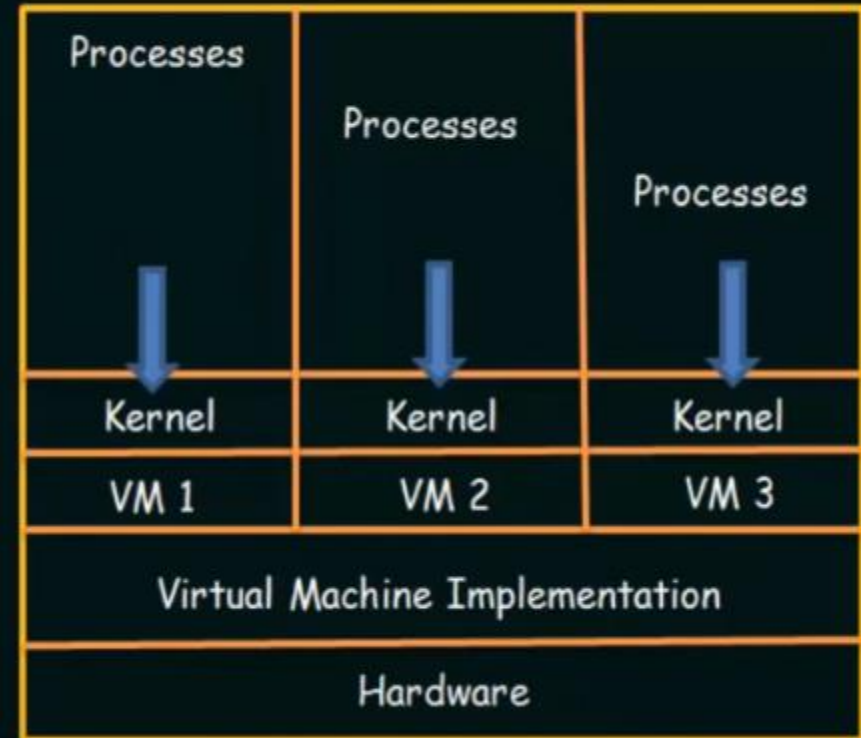
Runs in Kernel mode
Runs in User Mode

Just as the physical machine has two modes, however,
so must the virtual machine.

Consequently, we must have:

- A virtual user mode and
- A virtual kernel mode

BOTH OF WHICH RUN IN A PHYSICAL USER MODE



- 
- You can also try to install this virtual machine installation by installing VMware or Virtual Box virtual machine software.
 - So that you can have idea about different operating system in single system
 - Lets say if you have windows OS and after installing VMware or virtual box software or any other software, you can install another OS
- 